

Dataset Overview

The original dataset was collected from multiple sources, including bridge-mounted sensors, environmental monitoring stations, traffic monitoring systems, and corrosion inspection records. Raw sensor measurements provide real-time information about the bridge's operating conditions, surrounding environment, and structural response.

The primary sensor data include wind speed, wind direction, temperature, humidity, traffic volume, precipitation, and geographical information. Corrosion-related variables such as Time of Wetness (TOW), pH, chloride concentration, and annual corrosion rate were obtained from environmental monitoring and inspection records.

To enhance the predictive capability of the dataset, several feature engineering techniques were applied. New features representing bridge degradation, structural performance, and risk levels were generated from the raw measurements. These engineered features enable the development of machine learning and deep learning models for structural health monitoring, predictive maintenance, and digital twin applications.

Original Sensor Features

The following variables were directly collected from sensors, monitoring systems, or inspection databases:

- Date
- Mean_Wind_Speed
- Wind_Direction
- Average_Humidity
- Average_Temperature
- Minimum_Temperature
- MaximumTemperature
- Structure_Length
- Latitude
- Longitude
- BridgeAge
- AverageDailyTraffic
- TotalPrecipitation
- Mean_TOW
- Mean_PH
- Mean_Cl-
- Mean_Annual_Corrosion_Rate

Engineered Features

The following features were generated through feature engineering and domain-specific calculations:

CorrosionDepth

Estimated cumulative corrosion penetration calculated from corrosion exposure conditions and annual corrosion rates.

AnnualRate

Derived degradation growth rate used to quantify yearly material deterioration.

Wind-Induced Vibrations

Estimated structural vibration response generated from wind speed, bridge geometry, and environmental conditions.

Traffic Load

Calculated operational loading based on average daily traffic and bridge characteristics.

Structural Health

Composite structural health index representing the overall condition of the bridge. This index integrates environmental, corrosion, operational, and structural response factors.

Vibration Risk

Categorical risk level derived from wind-induced vibration measurements.

Corrosion Risk

Categorical risk level generated from corrosion-related indicators such as chloride concentration, pH, corrosion depth, and corrosion rate.

Traffic Load Percentage

Represents the percentage of traffic utilization relative to the bridge's designed traffic capacity.

Traffic Load Risk

Risk category generated from traffic load percentage and operational stress levels.

Structural Health Risk

Risk category generated from the structural health index and overall bridge condition assessment.

Dataset Processing Workflow

1. Raw data acquisition from bridge sensors and monitoring systems.
2. Data cleaning and preprocessing.
3. Missing value handling and quality control.
4. Feature engineering to derive structural and degradation indicators.

5. Risk categorization using engineering thresholds.
6. Preparation of regression and classification targets.
7. Development of machine learning, deep learning, and digital twin models.

This dataset supports Structural Health Monitoring (SHM), Predictive Maintenance, Explainable Artificial Intelligence (XAI), Digital Twin systems, and Multi-Task Deep Learning frameworks for suspension bridge infrastructure management.